

Highlights

An open-source GCR dosimetry pipeline for Mars mission risk assessment: organ-specific REID, SEP acute hazard, and global uncertainty quantification

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- Open-source, pip-installable pipeline for GCR dose and cancer risk (REID).
- Species composition constrained to real ACE/CRIS and PAMELA measurements.
- Validated against real MSL/RAD and 8 years of independent ISS telemetry.
- Value-of-information sweep ranks which uncertainty to reduce first.
- Launch timing alone, across two real historical solar cycles, shifts mission cancer risk by a factor of 3.4x.

An open-source GCR dosimetry pipeline for Mars mission risk assessment: organ-specific REID, SEP acute hazard, and global uncertainty quantification

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ABSTRACT

I describe GCRISK, an open-source Python package for estimating galactic cosmic ray (GCR) radiation exposure and cancer risk on crewed deep-space missions. GCR flux, generated via force-field solar modulation with inter-species composition constrained to ACE/CRIS and PAMELA measurements, is transported through slab shielding via the continuous-slowing-down approximation; organ-specific cancer risk (REID) is estimated across eleven organs. A single global scale is fit to the MSL/RAD cruise-phase absorbed dose (1.84 mGy/day at 16 g cm⁻² aluminum). Independent checks (shielding-depth dependence, material ratio, real MSL/RAD and ISS telemetry) agree within benchmark uncertainties; day-to-day correlation against real telemetry is confounded by a documented SEP event and is consistent with zero once it is excluded, but the multi-month trend is consistent, including against eight years of independent ISS data ($r = 0.62$). A Solar Energetic Particle module estimates acute organ dose for three historical events.

Global uncertainty is propagated via Latin Hypercube Sampling over nine physics and radiobiological parameters, giving a median REID of 6.5% for a 35-year-old male on a 259-day Earth–Mars transit at 16 g cm⁻² aluminum (p5–p95: 1.7–20.0%). Radiobiological parameters dominate REID variance (> 85%) over GCR physics (< 15%). A value-of-information sweep shows halving the ICRP-60 quality-factor uncertainty alone would cut p95 REID by 33.5%, more than every physics parameter combined. Sweeping launch date over two real solar cycles (2003–2022) shows median REID varies 3.4× from launch timing alone. The composition-constrained HZE fraction (13.0% at 16 g cm⁻² Al) is 1.7× the NSRL reference, attributed to the absence of nuclear fragmentation in the transport model; the pipeline's $Q_{\text{eff}} = 3.32$ is correspondingly 27% above the MSL/RAD measured value. Source code, tests, and validation data are freely available on GitHub.

1. Introduction

Deep-space human exploration beyond low-Earth orbit exposes crew members to galactic cosmic rays (GCR) and solar energetic particles (SEP) at fluence rates and energies far exceeding the shielding capacity of current crew vehicles. GCR is the chronic, unavoidable component of this exposure and sets the dose baseline for any multi-year mission, while SEP is episodic but can deliver an acute dose in hours; this paper's pipeline centers on the former, with a dedicated module for the latter (§3.5). The GCR environment at 1–1.5 AU consists of protons, helium, and heavier ions (the so-called HZE particles) spanning kinetic energies from a few hundred MeV/n to tens of GeV/n. Biological effectiveness peaks near the energy range where stopping power is highest. A round-trip crewed Mars mission (transit + surface stay) of ~900 days is estimated to deliver a total effective dose of the order 1–2 Sv (Zeitlin et al., 2013), which, depending on crew demographics and the risk model used, implies a radiation-induced cancer mortality probability that may approach or exceed NASA's career limit (NASA, 2014).

Quantifying that risk requires combining a calibrated, solar-modulation-responsive GCR spectral model, organ-resolved dose transport, a cancer risk model accounting for

HZE LET quality, and uncertainty analysis that separates reducible physics uncertainty from the larger radiobiological unknowns. Tools such as HZETRN (Wilson et al., 1991), OLTARIS, and Geant4/TOPAS each address subsets of these requirements with greater physical rigor, particularly nuclear fragmentation, but none is simultaneously openly available, pip-installable, and integrated into one uncertainty-quantified workflow. To my knowledge, there is no single open-source tool that combines GCR spectral modeling, organ-specific REID, SEP acute risk, and global uncertainty propagation in one reproducible, pip-installable workflow.

Here I present *gcrisk*, a Python package that, within the limitations in §4.5:

1. generates six-species GCR spectra with force-field modulation and ACE/CRIS/PAMELA-constrained composition, calibrated to MSL/RAD;
2. transports spectra through slab shielding via CSDA and nuclear attenuation with tabulated secondary neutrons;
3. computes organ-specific dose and dose equivalent at eleven ICRP-89 depths;
4. estimates REID via the Cucinotta (2013) EAR/ERR model;
5. quantifies SEP acute BFO dose for three canonical events; and

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6. propagates global uncertainty via 9-parameter Latin Hypercube Sampling with output variance decomposition.

The pipeline serves mission designers, educators, and researchers who need a transparent, reproducible, uncertainty-quantified *estimate* without standing up a closed high-fidelity code like HZETRN or Geant4. Physics parameters together contribute $\sim 12\%$ of REID variance; radiobiological parameters contribute $\sim 88\%$. The value-of-information and launch-window results below follow directly from that split, and are difficult to extract from existing closed tools.

2. Methods

All computation is implemented in Python 3.9+ as the open-source package `gcrisk`. The pipeline is a sequence of composable modules: spectrum generation, slab transport, organ dose, cancer risk (REID), SEP acute hazard, and uncertainty propagation, plus a separate proton RBE self-consistency check. Full equations, parameter tables, and derivations for every subsection below are given in Supplementary Methods (Sections S1.1–S1.9); only the essential physics and the numbers needed to interpret the Results are summarized here.

Trajectory and solar modulation. A Hohmann-transfer trajectory (Earth 1.0 AU to Mars 1.524 AU, 259-day transit) sets the mission timeline. Solar modulation potential $\phi(t)$ comes from the Usoskin et al. monthly reconstruction (real neutron-monitor data back to 1936); for the MSL transit epoch, $\phi \approx 481 \pm 50$ MV. Neutron-monitor inter-station offsets (± 50 – 100 MV) are propagated as an LHS parameter (§2.1).

GCR spectrum. Solar modulation of GCR flux uses the Gleeson–Axford force-field approximation applied to per-species local interstellar spectra: protons and helium come from Vos & Potgieter Vos and Potgieter (2015), and C/O/Si/Fe from the HelMod model of Boschini et al. Boschini et al. (2020). Inter-species flux ratios are constrained to ACE/CRIS (George et al., 2009) and PAMELA (Adriani et al., 2014) measurements at the MSL epoch. A single global scale is then fit to match the MSL/RAD absorbed dose of $1.84 \text{ mGy day}^{-1}$ at 16 g cm^{-2} aluminum (Zeitlin et al., 2013). This is a one-free-parameter calibration whose species mix is externally anchored, replacing an earlier six-free-parameter fit with no compositional constraint. The resulting HZE fraction (13.0% at 16 g cm^{-2} Al) is $\sim 1.7\times$ the NSRL reference field (Slaba, Blattnig, Norbury, Rusek and La Tessa, 2016), versus $\sim 4\times$ under the prior calibration; this residual excess is attributed to the CSDA transport model not capturing nuclear fragmentation (Supplementary §S1.3.3).

Transport. Ionization energy loss uses the continuous-slowing-down approximation (CSDA) with Barkas effective-charge scaling on NIST PSTAR/ASTAR proton stopping powers. Nuclear interactions are modeled as exponential fluence attenuation with a Bradt-Peters geometric mean free path (Bradt and Peters, 1950), the same form underlying HZETRN’s tabulated mean free paths (Wilson et al., 1991).

This does not track secondary fragment production, which is the model’s primary physical limitation, most relevant above $\sim 30 \text{ g cm}^{-2}$ for Fe/Si. Secondary neutron dose equivalent uses a pre-computed lookup calibrated to the HZETRN benchmark of Slaba et al. Slaba et al. (2014) (± 25 – 30% , via LHS).

Organ dose and quality factor. Absorbed dose and ICRP-60 dose equivalent (ICRP, 1991) are computed at 11 organ-specific tissue depths from ICRP Publication 89 (ICRP, 2002), weighted by ICRP-60 tissue factors. The field effective quality factor is $Q_{\text{eff}} = \sum H_{k,s} / \sum D_{k,s}$ (numeric values and the ion-vs.-neutron breakdown: §3.4).

Cancer risk (REID). REID is computed from the Cucinotta et al. Cucinotta, Kim and Chappell (2013) organ-specific ERR/EAR model (NASA/TP-2013-217375), using interpolated US sex-specific survival and baseline-mortality lookup tables as a practical approximation to the full NASA/Cucinotta framework. Point estimates use a fixed DDREF of 1.75; the uncertainty ensemble instead samples $\text{DDREF} \sim \text{Uniform}[1, 2]$. REID’s distribution is right-skewed because Q_{factor} (§2.1), the dominant tail driver, is drawn from a heavy-tailed lognormal; the effective quality factor is clipped at the ICRP-60 physical maximum of 30, and the resulting REID sample is separately clipped to $[0, 1]$ since it represents a probability (Supplementary §S1.6).

SEP acute risk. Event fluence spectra use the Band function (Tylka and Lee, 2006) for three canonical events (August 1972, October 2003, January 2005; parameters in Supplementary Table S3). BFO absorbed dose is computed via CSDA transport through wall plus 10 g cm^{-2} tissue and compared to the NASA 30-day BFO limit of 250 mGy (Cucinotta and Durante, 2010). Mission SEP encounter probability follows a Poisson process at solar-cycle-dependent rates (Feynman, Spitale, Wang and Gabriel, 1993).

2.1. Uncertainty Quantification

Nine parameters are varied simultaneously via Latin Hypercube Sampling (LHS): five physics (solar modulation, LIS normalization, nuclear cross-sections, neutron yield, HZE composition) and four biology (quality factor scale, DDREF, ERR scale, EAR scale), each with a lognormal, uniform, or clipped-normal prior justified by the relevant literature spread (full distributions and justifications in Supplementary Table S4). For each LHS sample, a full mission dose integration is performed; publication results use $N = 500$ samples. Parameter sensitivity uses normalized Spearman- ρ^2 decomposition (an approximation to first-order Sobol indices).

A separate value-of-information sweep (§3.8) re-runs the $N = 500$ ensemble once per parameter, halving that parameter’s prior spread while holding all others at baseline, to quantify how much narrowing any one source of uncertainty would actually reduce REID uncertainty (narrow argument to `gcrisk.uncertainty.lhs_samples`; Supplementary §S1.9).

2.2. Dose-Averaged LET and Proton RBE Self-Consistency Check

Per-species and field dose-averaged LET ($\bar{L}_D$) are computed to trace Q_{eff} to specific ion contributions (§3.4; full definitions in Supplementary §S1.9). The same $\bar{L}_D$ infrastructure also implements two proton RBE models, Wedenberg et al. (2013) (Wedenberg, Lind and Hardemark, 2013) and McNamara et al. (2015) (McNamara, Schuemann and Paganetti, 2015), matched to OpenTOPAS-RBE scorer conventions and checked only as an implementation self-consistency test (Table 1). This layer is the foundation for a separate, planned proton-therapy crossover study.

3. Results

3.1. Dose Rate Validation

Two independent checks benchmark the pipeline against data not used in calibration: the shielding curve shape stays within the HZETRN benchmark envelope ($\lesssim 20\%$) (Slaba et al., 2014) at 5, 10, and 30 g cm⁻² aluminum, and the Al/PE dose ratio (1.36) is consistent with Mrigakshi (2013)'s 1.43 ± 0.22 . As a separate internal self-consistency check, the pipeline's own dose rate correlates with its own ϕ input at $r = -0.998$ ($n = 260$ days).

For an independent daily comparison I obtained real MSL/RAD cruise-phase telemetry from NASA's PDS archive (MSL-M-RAD-3-RDR-V1.0; 218 of 253 cruise days; `scripts/fetch_real_rad_cruise_data.py`), replacing the statistically-reconstructed proxy used in earlier drafts. Day-to-day Pearson agreement over all 218 days is weak ($r = -0.127$, $p = 0.06$), but this is mostly an artifact of one real event: an X5.4 flare (AR11429) on 2012-03-07 produced one of the largest SEP events of solar cycle 24 (~ 6530 pfu), and RAD's raw dose rate spikes $\sim 30\times$ above baseline the next day. Since this pipeline models SEP as a physically separate, episodic module from chronic GCR (§3.5), comparing the GCR-only daily model against telemetry containing a live SEP spike *biases the comparison* for those few days.

Two independent checks confirm the weak r above is entirely the outlier's doing. *Spearman's ρ* (rank-based, insensitive to a single large spike) on the *full*, unmodified series is already ≈ 0 ($\rho = -0.007$, $p = 0.92$). And Pearson r computed with the documented event window excluded is also ≈ 0 ($r = -0.001$, $p = 0.99$, $n = 209$). Note that p *rising* here, from 0.06 to 0.99, is expected. A p-value answers "how consistent is this r with zero," and r itself moved from -0.127 toward 0, so p correctly moved toward 1. There is no cleaner signal hiding underneath the SEP day; removing it just removes the one thing that was pulling r away from zero.

I also tested whether giving the model genuine daily-resolution modulation would produce a real signal instead. That modulation was a real Oulu neutron-monitor daily count rate, calibrated to ϕ on an independent 1995–2011 window. It does not, $r = -0.127 \rightarrow -0.119$, ruling out " ϕ needs finer resolution" as an explanation too. Taken together,

this is the honest ceiling for a monthly-resolution force-field model against real telemetry that also carries its own counting-statistics noise: no daily-resolution signal, in either direction.

The multi-month trend the model *is* built to capture is consistent in magnitude and direction (real +17.5% vs. modeled +14.5%, first-to-last cruise decile). Because this RAD-anchoring is itself fit to the evaluated data, I also tested generalization by splitting the series in half and fitting/testing in both directions: held-out RMS error (21.4%, 24.8%) is close to in-sample RMS error (23.5%, 19.4%), a 5.5-point maximum degradation. This shows the anchoring generalizes beyond the period it was fit to.

Table 1 summarizes all comparisons by epistemic strength. The calibration-point dose rate (1.843 mGy day⁻¹ vs. RAD's 1.84 ± 0.33) is expected: it is what the single-point fit was tuned to match.

3.2. Independent Multi-Year Solar-Cycle Cross-Check

The MSL cruise window above is 253 days within a single solar cycle; as a longer, independent cross-check of the force-field modulation physics beyond that window, I compared the pipeline's phi-driven predicted dose-equivalent rate at 1 AU against real ISS radiation telemetry (ISS_DOSANL_TEPIC, NASA Space Radiation Analysis Group, via CDAWeb; `scripts/fetch_iss_tepic_data.py`, `scripts/compare_iss_solar_cycle.py`), May 2010–April 2018 — a full solar cycle, $\sim 12\times$ longer than the MSL window. Because the ISS orbits inside Earth's magnetosphere (geomagnetic shielding this pipeline does not model), only the modulation *trend* is testable here. Figure 1 shows both monthly-median series (91 months, one instrument-gap outlier dropped): $r = 0.617$ ($p = 7 \times 10^{-11}$), with both series declining toward the 2014 solar maximum and rising sharply through 2018.

3.3. Shielding Effectiveness and Uncertainty Bounds

Having validated the model's response to solar modulation, the next question is how added shielding mass actually converts into dose. Figure 2 shows absorbed dose rate vs. aluminum and polyethylene shielding (259-day transit, $\phi = 481$ MV) with p5–p95 LHS bands. Dose decreases steeply to ~ 10 g cm⁻², then flattens above ~ 16 –20 g cm⁻² as secondary-neutron buildup offsets HZE attenuation; the p5/p95 ratio at 16 g cm⁻² Al is ~ 1.9 (dominated by LIS normalization and neutron-yield uncertainty). Polyethylene gives $\sim 26\%$ lower dose than aluminum at 16 g cm⁻², consistent with hydrogen-rich materials fragmenting HZE ions more effectively.

The same shielding scan also shows which particles carry the dose at each depth, in Figure 3. Protons dominate throughout but their own fraction actually *falls* with depth (73.1% $\rightarrow$ 68.9% $\rightarrow$ 58.0% at 5/16/30 g cm⁻² Al), while carbon and oxygen's fractions *grow* (3.8% $\rightarrow$ 5.5% $\rightarrow$ 8.0% and 6.2% $\rightarrow$ 6.7% $\rightarrow$ 8.3%) even as silicon and iron are attenuated toward zero ($< 1.5\%$ by 5 g cm⁻², $< 0.1\%$

Table 1

Pipeline benchmarks grouped by epistemic status: *Independent* (not used in calibration), *Calibration* ($\dot{D}$ fitted; $\dot{H}$, Q_{eff} are predictions), *Self-consistency* (no external reference).

Quantity	Pipeline	Reference	Source	Rel. error
<i>Independent benchmarks (not used in calibration)</i>				
Shielding slope (5–30 g/cm ²)	within	HZETRN env.	Slaba et al. (2014)	< 20%
$\dot{D}$ ratio (Al vs PE, 16 g cm ⁻²)	1.36	1.43 ± 0.22	Mrigakshi et al. (2013)	5%
Cruise trend (early vs. late decile)	+14.5%	+17.5%	NASA PDS RAD RDR	3 pts
Daily $r(\dot{D}_{\text{model}}, \dot{D}_{\text{RAD}})$, $n = 218$	-0.127	—	NASA PDS RAD RDR	SEP-confounded, see §3.1
Daily r , SEP window excl., $n = 209$	-0.001	—	NASA PDS RAD RDR	$p = 0.99$
Monthly r , ISS solar-cycle, $n = 91$, 2010–18	0.617	—	ISS TEPC (CDAWeb)	$p = 7 \times 10^{-11}$
<i>Calibration point ($\dot{D}$ fitted; $\dot{H}$ and Q_{eff} are model predictions)</i>				
$\dot{D}$ (mGy/day) at 16 g cm ⁻² Al	1.84	1.84 ± 0.33	Zeitlin et al. (2013)	< 1%
$\dot{H}$ (mSv/day) at 16 g cm ⁻² Al	6.09	4.81 ± 0.97	Zeitlin et al. (2013)	+27%
Q_{eff} at 16 g cm ⁻² Al	3.32	2.62 ± 0.14	Zeitlin et al. (2013)	+27%
<i>Self-consistency (no external reference data)</i>				
RBE (V79, Wedenberg)	1.154	1.153	TOPAS-nBio	< 10 ⁻³
RBE (V79, McNamara)	1.165	1.165	TOPAS-nBio	< 5 × 10 ⁻⁸
Daily $r(\dot{D}, \phi)$, $n = 260$	-0.998	—	—	—

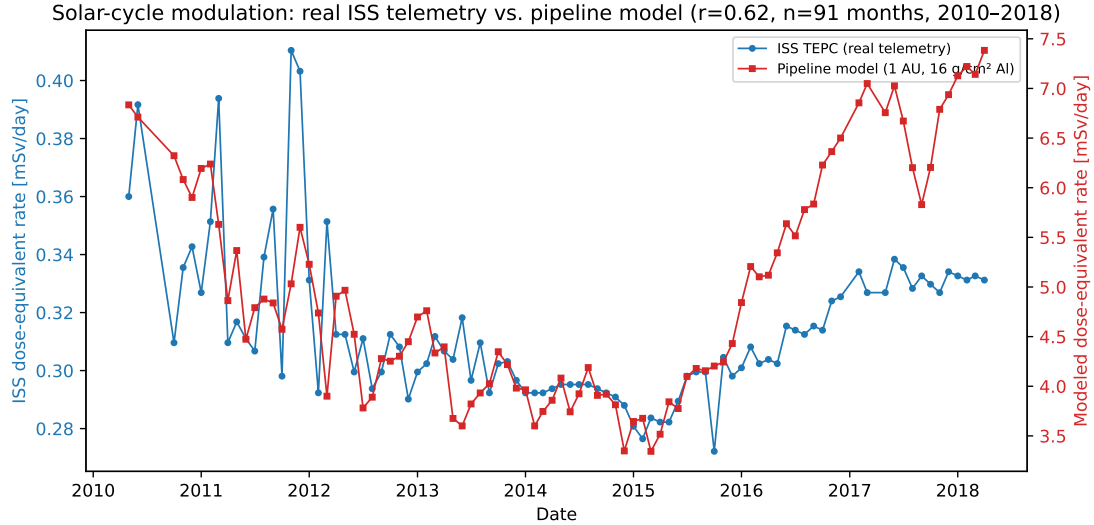


Figure 1: Real ISS TEPC telemetry (blue) vs. modeled dose-equivalent rate at 1 AU (red), May 2010–April 2018. Axes independently scaled (trend only, see text). $r = 0.617$, $n = 91$, $p = 7 \times 10^{-11}$.

by 30 g cm⁻²). Since this transport model does not track nuclear fragmentation (§4.5), this shift reflects each species' own CSDA range-energy relationship playing out under increasing shielding: protons' steep, low-energy-dominated spectrum is preferentially stopped as more material is added, while carbon and oxygen's harder spectra let a larger share of their own dose survive to greater depth.

3.4. Per-Species LET Decomposition

The plateau above ~ 16 g cm⁻² reflects a shift in which species carry the dose. Figure 4 shows dose-averaged LET per species and fractional dose contributions at 0, 16, and 30 g cm⁻² aluminum. Unshielded at $\phi = 481$ MV, the HZE group ($Z > 2$) is 20.9% of absorbed dose (oxygen 8.8%, iron 5.8% dominant), consistent with GCR abundance ordering

and Z^2 stopping-power scaling; field $\bar{L}_D = 3.9$ keV μm^{-1} gives $Q_{\text{eff}}^{\text{ions}} = 3.30$. After 16 g cm⁻² aluminum the HZE fraction falls to 13.0% (protons rise to 68.9%), field $\bar{L}_D$ drops to ~ 2.5 keV μm^{-1} , and the neutron contribution drives the total apparent $Q_{\text{eff}} = 3.32$ (§2).

3.5. SEP Acute Risk

Chronic GCR exposure is only half the radiation picture; episodic SEP events pose a distinct, acute hazard on the timescale of hours rather than years. Figure 5 shows BFO absorbed dose from three canonical SEP events vs. aluminum shielding. The August 1972 event delivers $\sim 37,000$ mGy unshielded ($\sim 148\times$ the 30-day BFO limit of 250 mGy) and stays above the limit until ~ 32 g cm⁻²; October 2003

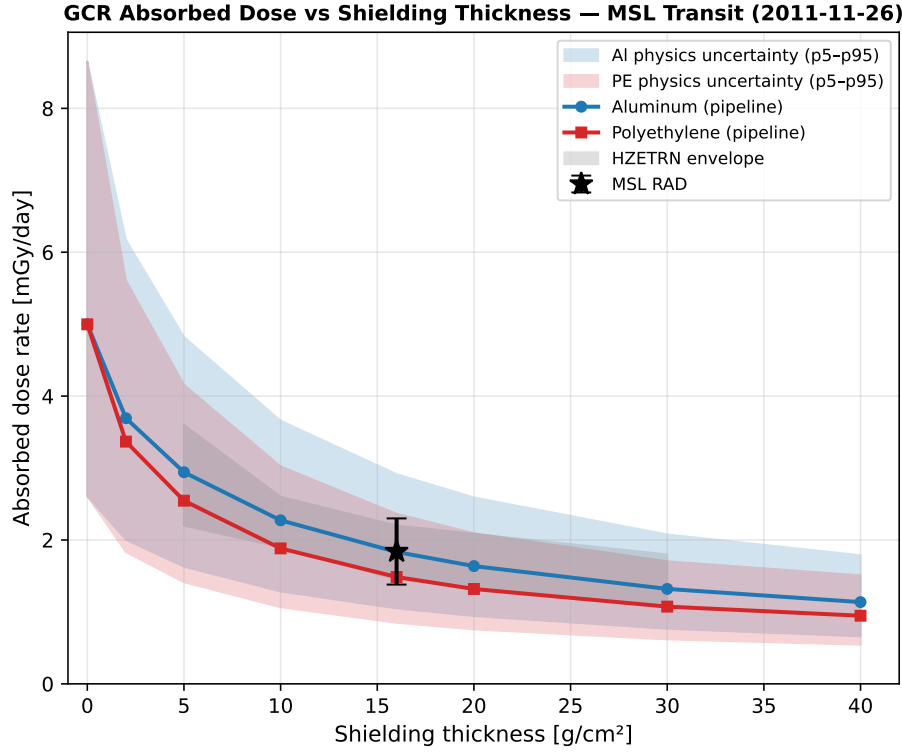


Figure 2: Dose rate vs. Al/PE shielding with p5–p95 bands; MSL/RAD point (star); HZETRN envelope (gray).

needs $\sim 25 \text{ g cm}^{-2}$; January 2005 falls below the limit under 20 g cm^{-2} . GCR and SEP respond very differently to shielding: above $\sim 20 \text{ g cm}^{-2}$, added aluminum barely reduces chronic GCR dose but still cuts SEP dose sharply. The August 1972 spectrum predates space-borne dosimetry (reconstructed from ground-level neutron-monitor/riometry data (Tylka and Lee, 2006)) and should be read as an order-of-magnitude estimate ($\sim \pm 50\%$).

3.6. REID vs. Shielding and Organ Contributions

Returning to the chronic GCR side, mission planners ultimately need cancer risk. Figure 6 shows REID vs. aluminum shielding for a 35-year-old male and female on a 259-day transit. At 16 g cm^{-2} , median REID is 6.5% (male) and 8.3% (female), both above the historical NASA 3% career limit (NASA, 2014). The female excess reflects higher sex-specific lung/breast ERR-EAR coefficients (Cucinotta et al., 2013). p5–p95 bands (1.7–20.0% male, 2.2–25.7% female) reflect the dominant radiobiological uncertainty (§2.1): the mission carries meaningful cancer risk at a precision the current radiobiological framework cannot resolve. Stomach, colon, and lung contribute most to organ dose equivalent, consistent with their high ICRP-60 weighting ($w_T = 0.12$).

3.7. REID vs. Launch Date Across the Solar Cycle

REID depends strongly on the solar modulation potential ϕ at launch. I swept `gcrisk.reid.reid_vs_launch_date` over monthly launch dates (`scripts/launch_window_sweep.py`); Figure 7 separates two regions. The solid curve (2003–2022) uses real Usoskin ϕ data spanning two full solar

cycles; median REID ranges from 3.3% (May 2003, solar maximum, $\phi \approx 853 \text{ MV}$) to 11.3% (May 2009, solar minimum, $\phi \approx 388 \text{ MV}$). That is a $3.4\times$ spread from launch timing alone, at fixed shielding and demographics. The dashed, hatched curve (2023–2035) is a self-generated idealized ~ 11 -year cosine cycle, included only to illustrate the qualitative shape a tool like this could show for a future window *if* the cycle repeats its characteristic period. It is illustrative only, stopping short of a space-weather forecast or a recommended launch date. Real cycles vary in amplitude and length by years; an operational decision would need an actual forecast, outside this paper's scope.

3.8. Value of Narrowing Individual Uncertainty Parameters

Launch timing is a free lever available at zero engineering cost; tightening a physical or biological parameter is not, so the natural next question is which parameter is worth that investment. The variance decomposition (§2.1) ranks parameters by their share of REID variance under *current* priors, but not how much narrowing any one would help. I re-ran the $N = 500$ ensemble once per parameter, halving its uncertainty while holding the rest at baseline (`scripts/sensitivity_narrowing_sweep.py`); Table 2 ranks the resulting p95 REID reduction. Halving Q_{factor} alone shrinks p95 REID from 20.0% to 13.3% (33.5% reduction) — $\sim 7\times$ more than the next parameter (DDREF, 4.5%) and more than every GCR physics parameter combined ($< 5\%$). This makes *biology dominates* concrete: tightening the ICRP-60

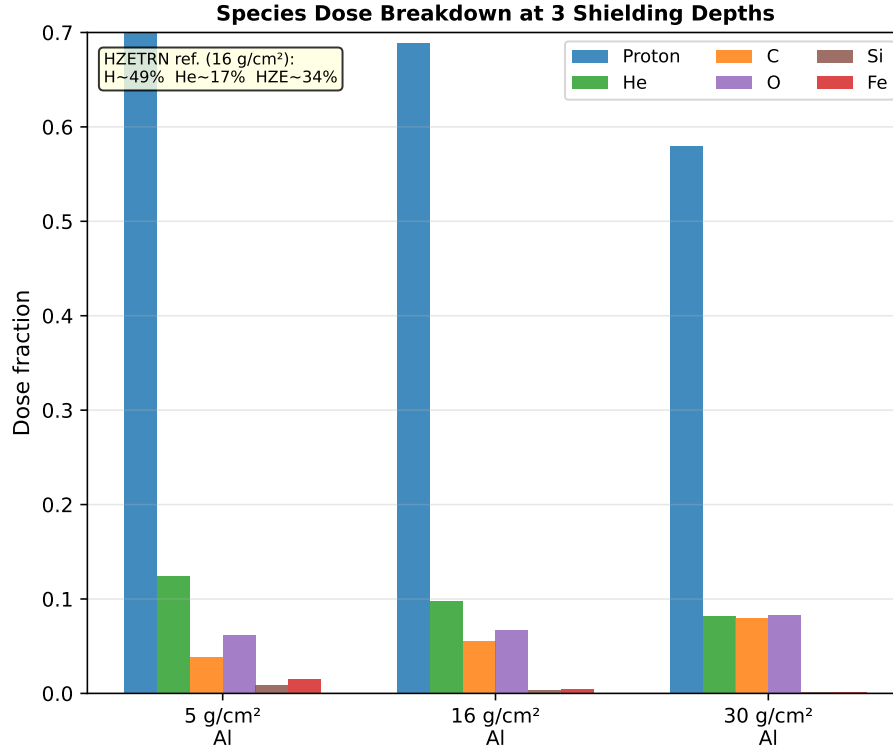


Figure 3: Per-species dose fractions at 5, 16, 30 g cm⁻² Al, with the HZETRN reference field's fractions at 16 g cm⁻² annotated for comparison.

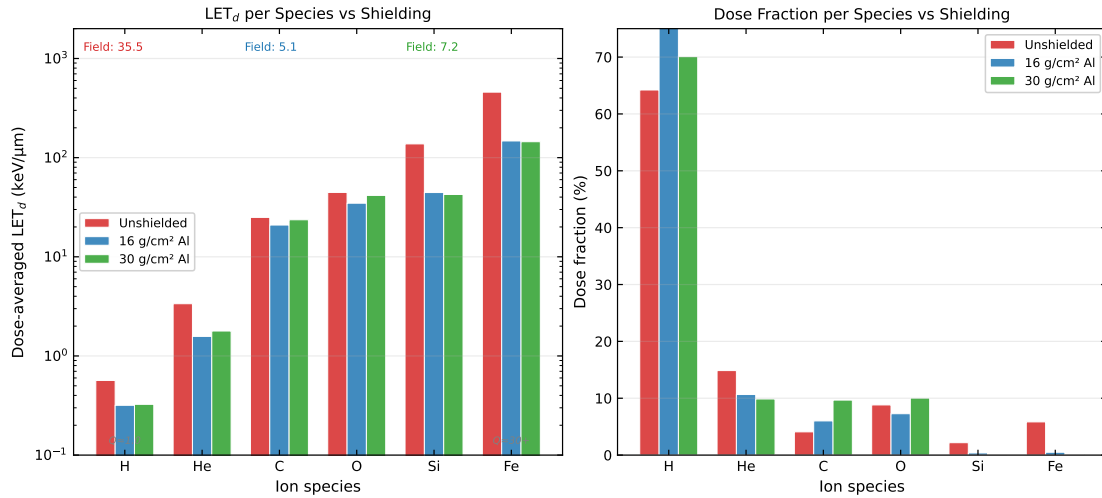


Figure 4: Left: dose-averaged LET per species at 0, 16, 30 g/cm² Al. Right: fractional dose contribution per species; HZE fraction drops 20.9% → 13.0% as heavier ions attenuate preferentially.

quality-factor-model uncertainty (e.g. expert elicitation or a track-structure-resolved $Q(L)$ for mixed GCR fields) would narrow Mars mission REID uncertainty more than every other parameter here combined.

4. Discussion

4.1. What the Validation Does and Does Not Show

The independent checks in §3.1–3.2 (shielding curve shape, material ratio, the real MSL/RAD cruise trend, and the real ISS multi-year trend) each probe an aspect of the model not constrained by the single-point MSL/RAD calibration. Together they support characterizing the pipeline as calibrated to one flight measurement and consistent with

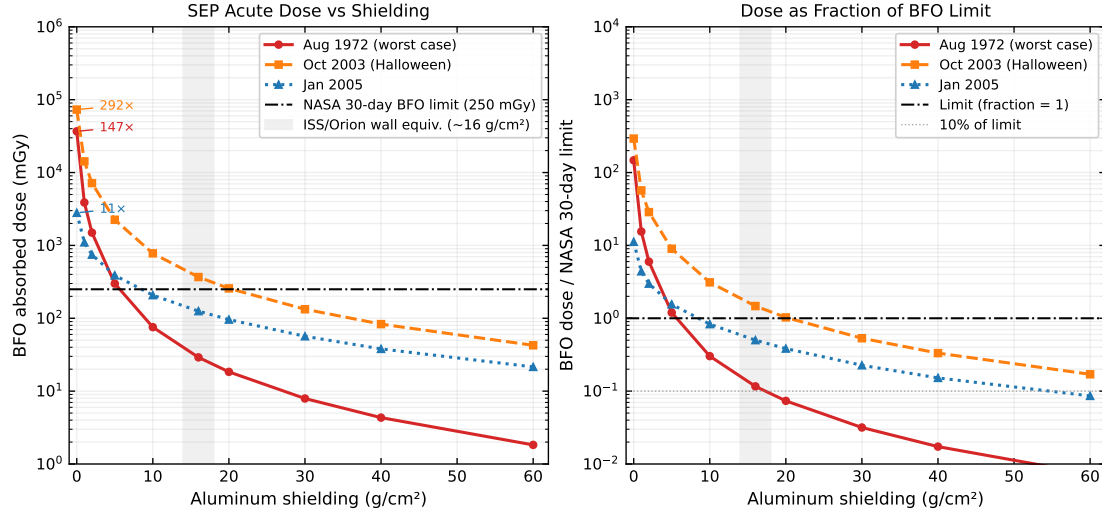


Figure 5: Left: BFO dose (mGy) vs. Al shielding; line = NASA 30-day limit (250 mGy); band = ISS/Orion wall equivalent. Right: normalized to limit.

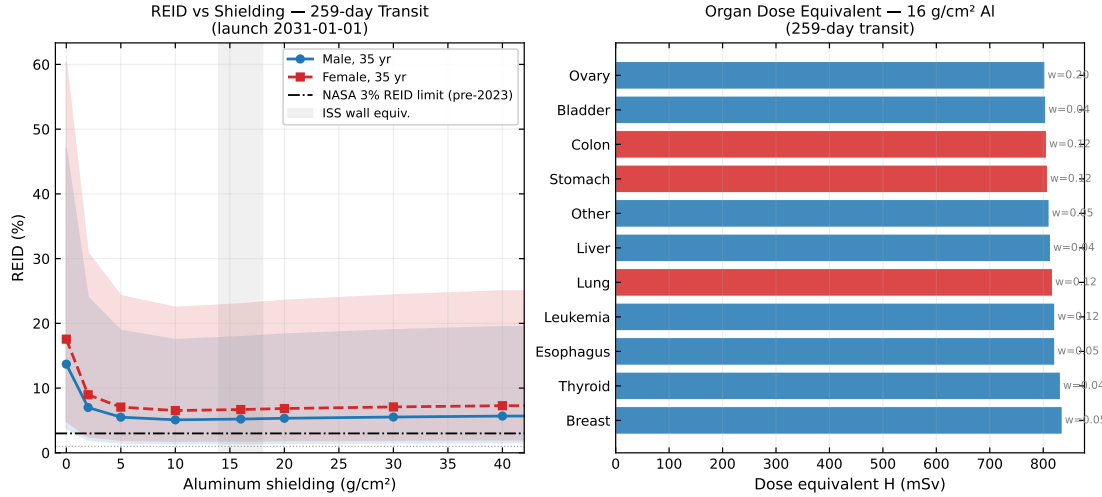


Figure 6: Left: REID (%) vs. Al shielding, male/female, p5–p95 bands; line = NASA 3% limit. Individual samples are capped at 100% (REID is a probability); the wide band below $\sim 5 \text{ g cm}^{-2}$ reflects genuinely large compounding radiobiological uncertainty at high unshielded dose, not an artifact. Right: organ dose equivalent at 16 g cm^{-2} Al, w_T annotated.

independent benchmarks within their stated uncertainties. The near-zero daily-resolution correlation, the SEP-event confound behind it, and the held-out cross-validation of the RAD anchoring (§3.1) are reported for the same reason: an honest, non-circular account of what this monthly-resolution model can and cannot claim is worth more than an artificially strong number. That discipline paid off here: chasing the daily correlation initially suggested a resolution gap in the solar-modulation physics, but a genuine daily-resolution reconstruction (real neutron-monitor data, independently calibrated) did not improve it, and the actual explanation turned out to be a documented SEP event contaminating the comparison for a few days.

4.2. Implications for Mission Shielding Design

Turning from validation to what these numbers imply operationally, the shielding curve (Fig. 2) plateaus above $\sim 20 \text{ g cm}^{-2}$ aluminum as secondary-neutron growth offsets HZE attenuation: going from 16 to 40 g cm^{-2} cuts GCR dose only $\sim 42\%$ while roughly doubling wall mass, consistent with prior literature (Cucinotta, Kim and Ren, 2005; Slaba et al., 2014) that bulk aluminum shielding is mass-inefficient against chronic GCR. SEP shielding is the opposite case: the October 2003 event drops from $\sim 35\times$ the BFO limit unshielded to below it by $\sim 25 \text{ g cm}^{-2}$ (Fig. 5). This asymmetry supports a compact, heavily-shielded SEP storm shelter over whole-vehicle HZE shielding as the more mass-efficient architecture (Durante and Cucinotta, 2011; Cucinotta and Durante, 2010).

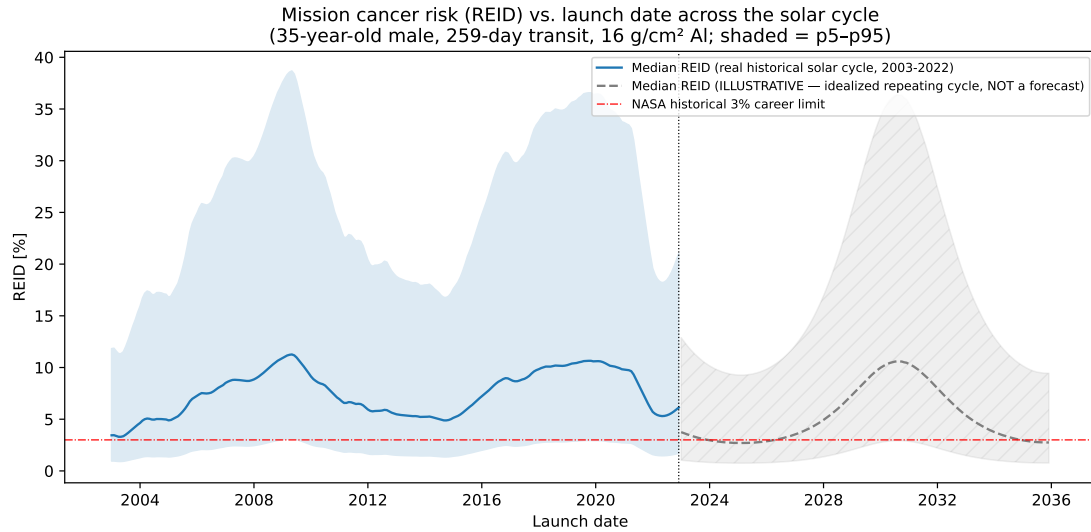


Figure 7: REID vs. launch date (male, 16 g cm^{-2} Al; shaded = p5–p95). Solid blue: real historical ϕ (2003–2022). Dashed/hatched: an idealized repeating cycle, *not* a forecast (see text). Line = NASA 3% limit.

Table 2

Value-of-information sweep: reduction in p95 REID from halving one parameter's uncertainty at a time (35-year-old male, 259-day transit, 16 g cm^{-2} Al; baseline p95 = 20.0%).

Parameter	Narrowed p95	Δ p95	% of baseline
Q_factor	13.3%	−6.7 pts	−33.5%
DDREF	19.1%	−0.9 pts	−4.5%
phi_scale	19.5%	−0.5 pts	−2.7%
ERR_scale	19.8%	−0.3 pts	−1.3%
LIS_norm	19.8%	−0.2 pts	−1.1%
hze_norm_scale	19.9%	−0.2 pts	−0.8%
neutron_H	20.0%	−0.1 pts	−0.3%
EAR_scale	20.0%	−0.0 pts	−0.1%
cross_sec	20.0%	0.0 pts	0.0%

4.3. REID Interpretation: Biology Dominates

The shielding trade-offs above are only as meaningful as the risk number they are designed around, so it is worth scrutinizing that number directly. Median REID (6.5% male, 8.3% female at 16 g cm^{-2}) exceeds the historical NASA 3% limit (NASA, 2014), though NASA's updated age/sex-dependent framework (NASA, 2023) is not implemented here and the comparison should be read as indicative. Because the composition-constrained pipeline overestimates Q_{eff} by $\sim 27\%$ (§2), these values are conservative upper estimates; correcting for that ratio would put male median REID at $\sim 5.1\%$ — still above the historical limit. The p5–p95 band spans a factor of $\sim 12\times$: this REID should be read as "the mission dose likely carries meaningful cancer risk, with the exact value uncertain by an order of magnitude."

Most importantly, this uncertainty is overwhelmingly biological. The four radiobiological parameters contribute $\gtrsim 85\%$ of REID's normalized sensitivity, all five physics parameters combined $< 15\%$ (Supplementary Table S4).

This is consistent with Cucinotta et al. (2013) and the broader literature (Durante and Cucinotta, 2011). No direct human data constrain the HZE LET range that dominates unshielded GCR dose equivalent; extrapolating from mouse and atomic-bomb-survivor data to this regime is the single largest source of uncertainty in this calculation, and no amount of improved transport physics can resolve it. Strikingly, ϕ_{scale} drives 79% of absorbed-dose variance but only 8% of REID variance: solar-modulation uncertainty strongly affects physical dose but has almost no effect on mortality probability once biology enters. The value-of-information sweep (§3.8) makes this actionable rather than qualitative.

4.4. Scope of the LET/RBE Layer

Separately from the REID interpretation above, it is worth being explicit about what the pipeline's LET/RBE layer does and does not claim. The per-species LET decomposition (Fig. 4) traces Q_{eff} 's physical origin to specific ion contributions; the residual 27% Q_{eff} excess over the RAD measurement is attributed to CSDA's missing fragmentation (§2). The repository also contains a proton RBE self-consistency layer (§2.2) reported only as an implementation check (Table 1); it is the foundation for a planned, separate proton-therapy crossover study.

4.5. Limitations

The main structural limitation is CSDA transport without nuclear fragmentation, which by my order-of-magnitude estimate overestimates Q_{eff} (and correspondingly REID) at 16 g cm^{-2} by roughly 15–25% (Wilson et al., 1991; Slaba et al., 2014). That is well within the $12\times$ uncertainty band but not negligible relative to the median. Other major approximations include: a uniform-slab spacecraft geometry (no ray-tracing); six modeled GCR species (unmodeled species

< 5% of dose); the force-field modulation's monthly resolution, which limits daily-scale claims and is the reason the ISS cross-check is restricted to trend and does not extend to near-Earth or trapped-radiation environments; the pre-2023 Cucinotta NASA risk framework rather than the updated age/sex-dependent limits; and cancer-only risk (no CNS, cataract, or cardiovascular endpoints). The full itemized discussion of each limitation, with supporting estimates, is provided in Supplementary §S2.

4.6. Future Work

Each limitation above points to a concrete next step, listed roughly in order of expected impact on REID.

- **Nuclear fragmentation:** replacing the Bradt-Peters fluence-removal model with tabulated fragmentation cross-sections (Wilson et al., 1991) would let secondary light-ion production be tracked explicitly, closing the 15–25% Q_{eff} overestimate (§4.5) rather than only bounding it.
- **Risk framework:** adopting NASA's 2023 age/sex-dependent career-limit framework (NASA, 2023) in place of the pre-2023 Cucinotta model would remove the main caveat on the REID-vs-NASA-limit comparison in §4.3.
- **Geometry:** ray-traced, non-uniform spacecraft shielding (as in OLTARIS or Geant4) would replace the uniform-slab approximation, most relevant for real vehicle interiors with mass concentrated unevenly around the crew.
- **Non-cancer endpoints:** CNS, cataract, and cardiovascular risk models are not yet implemented; CNS effects in particular have rodent-study support for a cognitive-impairment threshold below the cancer risk threshold modeled here.
- **Proton RBE:** the dose-averaged LET/RBE self-consistency layer (§2.2) exists as the foundation for a planned, separate proton-therapy crossover study.

4.7. Comparison with Existing Tools and Reproducibility

Set against the alternatives, Table 3 compares this pipeline against HZETRN, OLTARIS, Geant4, and TOPAS: it trades high-fidelity transport physics for being, to my knowledge, the only one that is simultaneously open/pip-installable and includes organ-specific REID, an SEP library, and LHS uncertainty quantification.

All results reproduce via `pip install -e .[dev], validate_pipeline.py, validate_extended.py, and pytest`; the full $N = 500$ ensemble takes ~ 90 minutes single-core, with a reduced $N = 20$ path used in CI. No proprietary datasets are required.

5. Conclusion

I have presented GCRISK, an open-source, pip-installable Python package integrating GCR spectrum generation, CSDA

Table 3

Comparison of GCR dosimetry tools. ✓ = present, △ = partial, × = absent.

Feature	Ours	HZETRN	OLTARIS	Geant4	TOPAS
Open / pip-install	✓	×	×	✓	✓
Fragmentation	×	✓	✓	✓	✓
Organ REID	✓	△	✓	×	×
SEP library	✓	△	✓	×	×
LHS uncertainty	✓	×	×	×	×
Clinical RBE	✓	×	×	△	✓
CI test suite	✓	×	×	×	×

transport, organ-specific REID, SEP acute dosimetry, and LHS uncertainty propagation, calibrated to MSL/RAD and consistent with HZETRN benchmarks within stated uncertainties. Median REID at 16 g cm^{-2} Al on a 259-day transit is 6.5% (p5–p95 1.7–20.0%), above the NASA 3% limit; the wide band is dominated by biological uncertainty. On the acute side, the August 1972 SEP event requires $> 30 \text{ g cm}^{-2}$ Al for sub-limit BFO protection, and at the 16 g cm^{-2} reference geometry its dose is $\sim 4\times$ the limit. The HZE group ($Z > 2$) accounts for 20.9% of unshielded dose ($\bar{L}_D = 3.9 \text{ keV } \mu\text{m}^{-1}$), falling to 13.0% at 16 g cm^{-2} Al, and the model's solar-modulation trend independently tracks 8 years of real ISS telemetry across a full solar cycle ($r = 0.62$, $p = 7 \times 10^{-11}$, $n = 91$). This is a longer, separately-sourced cross-check than the MSL calibration window it was fit to. Two results point toward what would most improve the estimate: halving Q_{factor} uncertainty alone cuts p95 REID by 33.5%, $\sim 7\times$ more than any other parameter or all physics parameters combined (§3.8), identifying where research investment matters most; and launch timing alone, over two real historical solar cycles (2003–2022), swings median REID $3.4\times$ (§3.7) at zero engineering cost.

All source code, 86 automated tests, and validation datasets are available at <https://github.com/aryanhshah8/gcrisk>.

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Declaration of Competing Interests

The author declares no competing interests.

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Data Availability Statement

All source code, input datasets, validation scripts, and TOPAS benchmark geometry files are freely available at <https://github.com/aryanhshah8/gcrisk>. Full methods derivations, extended parameter tables, and extended limitations discussion are provided as Supplementary Material. No proprietary datasets were used; all input data (Usoskin solar modulation potential database, NIST stopping powers, real MSL/RAD cruise-phase dosimetry from NASA's PDS archive, and real ISS TEPC telemetry from NASA's CDAWeb) are included in the repository under `data/`, and the HZETRN neutron benchmark table is hardcoded as a lookup array in `gcrisk/neutron_table.py`. Results can be fully reproduced without any proprietary software.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude (Anthropic) to:

- organize project files and directory structure;
- review and assist in debugging the Python pipeline code;
- assist in the writing and revision of the manuscript.

After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRedit authorship contribution statement

Aryan Hemendra Shah: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

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